

Who pays for robotics labs, and why students don't

One of the most common questions about industry-academia robotics programmes concerns funding: Who funds these labs, and are students expected to pay? Across the automation industry, the dominant model is a shared-investment partnership rather than a fee-based training system.

1. Industry partners (robot manufacturers, automation companies, solution providers) typically invest in:
 - Industrial robots (SCARA, 6-axis, controllers)
 - Software licenses, simulation tools, and safety systems
 - Initial faculty training and technical support
2. Academic institutions contribute:
 - Physical lab space and infrastructure
 - Faculty coordination and curriculum alignment
 - Long-term maintenance support

Crucially, students are not charged to access these labs. The objective is workforce readiness. Companies benefit indirectly by developing a talent pool already familiar with industrial platforms, thereby reducing onboarding time for graduates entering manufacturing roles. The approach is increasingly viewed as a strategic investment rather than a marketing or CSR activity.

efforts fit into this larger transition, serving less as standalone programmes and more as markers of a wider transformation already underway.

What is changing in robotics right now?

Not long ago, industrial robots were evaluated with simple metrics: payload, reach, and cycle time. If those numbers were strong, the robot was considered good. But what happens when a part is slightly misaligned, a product variant changes, or a line must be reconfigured within months? That is where today's robotics story begins.

Robots are no longer expected to simply repeat a pre-taught motion in a controlled environment. Today, they are gradually expected to sense, interpret, and adapt. Intelligence, software integration, and connectivity have become not just favourable but essential. This is a paradigm shift that is fundamentally altering the definition of 'performance.' Accuracy is still important. Speed is still important. But so is the ability to interface with vision systems, handle variability, and integrate with digital manufacturing systems,

rather than working alone.

As robots become smarter, they are also becoming smaller. The answer lies in broader changes across factories themselves. Modern manufacturing is less about running a single product for years and more about frequent changeovers and shorter production cycles.

High-speed selective compliance assembly robot arm or articulated robot arm (SCARA) robots clearly illustrate this trend. Systems such as Epson's LS-, G-, and T-Series, Yamaha Robotics' YK-X series, FANUC's SR SCARA robots, and IAI's tabletop SCARA platforms are now common in electronics and precision assembly. Their strength lies not in brute force but in repeatable accuracy, fast motion, and efficient use of limited floor space. For many manufacturers, compact SCARA robots paired with vision and software-driven control offer the flexibility needed to handle multiple product variants without constant mechanical redesign.

Not every task, however, can be addressed with planar motion alone. When parts must be approached from different angles or handled in confined spaces, full articulation becomes necessary. This is where compact six-axis

robots come in. Models such as ABB's IRB 1100 and IRB 1200, FANUC's LR Mate series, KUKA's KR Agilus, Yaskawa's GP-Series, and Epson's VT- and N-Series are increasingly deployed in applications requiring flexibility without the size or energy consumption of traditional industrial arms.

What stands out is not only their mechanical capability but how they are used. These robots are rarely standalone. They are typically integrated with vision systems, force control, and data connectivity, enabling them to respond to variation and feed production data upstream. The question is no longer whether a robot can move parts, but whether it contributes to process intelligence.

The rise of both SCARA and six-axis platforms signals a broader shift in manufacturing strategy. Long, rigid automation lines are giving way to smaller, reconfigurable production cells.

Compact robots fit naturally within this approach. They are easier to redeploy, quicker to integrate, and closely aligned with software-driven production planning. Rather than building automation for a single product, companies are building automation for constant change.

Robotics today is not defined by form factor alone but by expectations. Machines are judged not only on precision and repeatability, but also on adaptability, connectivity, and their ability to function within digital ecosystems. As intelligence becomes central to production, robotics is evolving from a purely mechanical discipline to a system-level one. The adoption of compact SCARA and articulated robots reflects how modern factories operate and evolve.

Why AI is moving to the edge

Robots on today's factory floors no longer follow rigid, pre-programmed instructions alone. One of the most